



TEAM VOID • HACKATHON 2026

TEAM ID: 2392608277

NEXT-GEN ACADEMIC LOGISTICS

SeatPlan AI

Exam seating with zero clashes, replanned in milliseconds.

 Zero-Latency Heuristic Engine

 Automated Anti-Cheating Layouts

 100% Client-Side Architecture

THE CORE CHALLENGE

Manual Planning Chaos in Exam Halls

48+ Hours of Spreadsheets

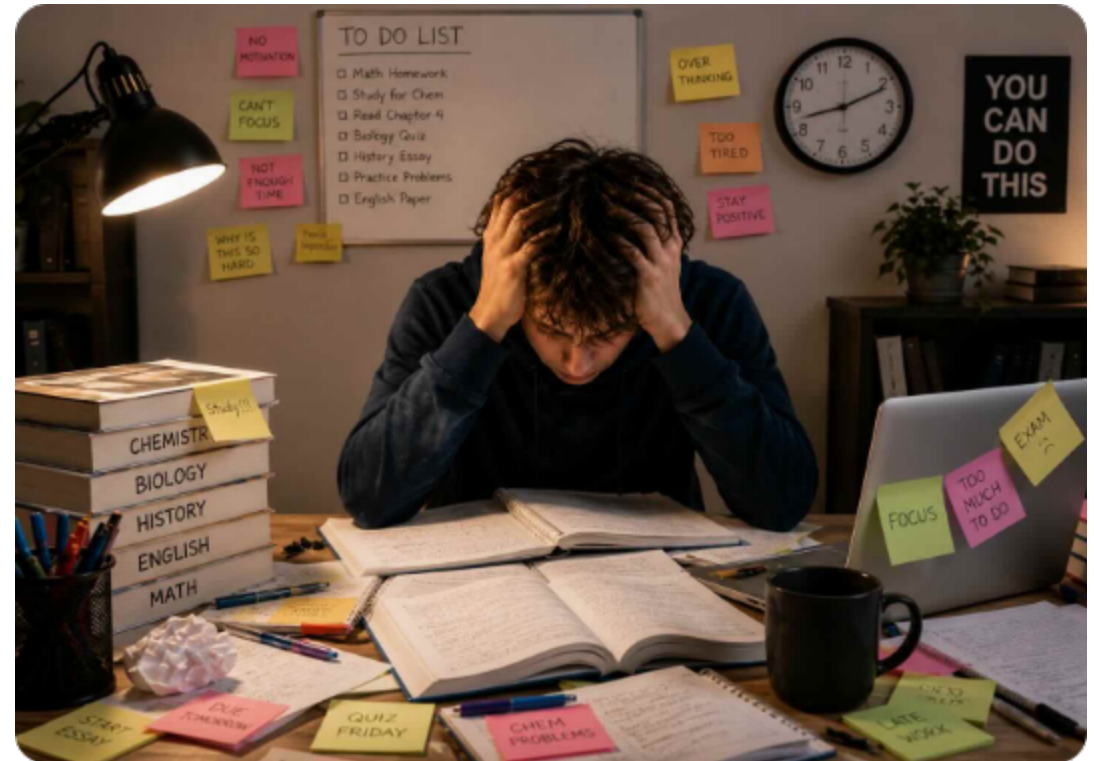
Staff manually juggle hundreds of roll numbers and rooms under intense deadline pressure.

Hidden Seating Clashes

Same-subject students placed side-by-side go unnoticed until test morning.

Last-Minute Hall Dropouts

When a room or teacher cancels, coordinators must re-chart hundreds of desks by hand.



Who Suffers From Flawed Seating?



Coordinators

Days lost to manual tables.
Constant stress and zero
margin for error.



Students

Last-minute relocations, room
confusion, and compromised
exam integrity.



Invigilators

Double-booked rosters and
unclear ratios in overcrowded
halls.



Institutions

Accreditation risks, student
disputes, and costly re-
examinations.

Deterministic Speed & Accuracy

0

Adjacent Clashes

Mathematical guarantee across all 4 Cartesian vectors.

< 50ms

Re-Optimization Time

Sub-second reallocation when halls or proctors cancel.

98%

Admin Hours Saved

Converts multi-day manual drudgery into a single click.



Built for Hackathon Real-World Readiness: Handles arbitrary room grids, mismatched cohort sizes, and dynamic room exclusions without database crashes.

Algorithmic Constraint Solving

1

Mix Subjects

Interleaves varied curricula in each examination room.

2

No Neighbours

Enforces Cartesian buffers: no identical peers left, right, front, or back.

3

Spare Buffer

Leaves vacant buffer desks rather than compromising integrity rules.

4

Smart Proctors

Maintains 1:30 invigilator ratios with strict zero double-bookings.

Client-Side Autonomous Workflow

PHASE 1



Generate & Map

Instantly synthesizes students, rooms, and proctors into spatial seat matrices.

PHASE 2



Verify & Replan

Separate verification routine recalculates constraints upon hall or staff removal.

PHASE 3



Zero-Leak Execution

100% in-browser offline calculation ensures zero student data leakage.

Legacy Tools vs SeatPlan AI

Capability	Manual Spreadsheets	Generic ERP Schedulers	SeatPlan AI
Generation Speed	2 - 3 Days	2 - 4 Hours	< 1 Second
Adjacency Anti-Cheating	Manual spot checks	Basic row alternate	4-Way Full Cartesian
Dynamic Dropout Handling	Rebuild from scratch	Complex re-run queue	Instant Minimal Move
Infrastructure Required	Excel / Google Sheets	Expensive Cloud Server	Zero Install (Any Web Browser)

From Hackathon MVP to Campus Production



Production CSV & SIS Integration

Direct sync with institutional databases for automatic cohort imports and printable seat manifests.



Multi-Slot Shift Management

Cross-session scheduling with faculty duty fatigue limits and morning-evening shift balancing.



Student QR Desk Verification

Students scan phone at exam hall door to see live desk number, instantly updated on reassignments.



Meet Team Void

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Thank You!

SeatPlan AI: Turning hours of exam panic into instant algorithmic certainty.

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Ready for live demo & questions from the jury.